

Hypothesis Testing and A/B Testing

Important Concepts

1. Null Hypothesis (H_0)

The **null hypothesis** states that there is **no significant difference or no effect** between the two groups or versions being compared.

It represents the default assumption that the new change has not produced an improvement.

2. Alternative Hypothesis (H_1)

The **alternative hypothesis** states that there **is a significant difference or effect** between the two groups.

When the question specifically asks whether something **increases** a metric, the alternative hypothesis is usually written as a **one-tailed hypothesis**.

3. Significance Level (α)

The significance level is the maximum probability of rejecting a true null hypothesis.

A commonly used significance level is:

$$\alpha = 0.05 = 5\%$$

4. p-value

The **p-value** helps us decide whether the observed result is statistically significant.

Decision rule:

- If **p-value** $\leq \alpha$, reject H_0 .
 - If **p-value** $> \alpha$, do not reject H_0 .
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Question 1

Scenario

Zomato wants to know if a new restaurant listing design increases the average number of daily orders compared to the old design.

Step 1: Define the Variables

Let:

- $\mu_{\square}$ = average daily orders with the **new design**
- $\mu_{\circ}$ = average daily orders with the **old design**

The question asks whether the new design **increases** average daily orders.

Step 2: State the Hypotheses

Null Hypothesis (H_0):

$$H_0: \mu_{\square} \leq \mu_{\circ}$$

The new restaurant listing design does not increase the average number of daily orders.

Alternative Hypothesis (H_1):

$$H_1: \mu_{\square} > \mu_{\circ}$$

The new restaurant listing design increases the average number of daily orders.

Interpretation

This is a **one-tailed test** because Zomato is specifically interested in whether the new design produces **higher** daily orders.

Final Answer

Hypotheses	Statement
H_0	The new design does not increase average daily orders.
H_1	The new design increases average daily orders.

Question 2

Scenario

A/B testing is performed on two versions of a Flipkart checkout page. The obtained p-value is **0.03**.

The significance level is:

$$\alpha = 0.05$$

Step 1: Compare p-value with significance level

Given:

$$\text{p-value} = 0.03$$

$$\alpha = 0.05$$

Therefore:

$$0.03 < 0.05$$

Step 2: Apply the Decision Rule

Since:

$$\text{p-value} \leq \alpha$$

we **reject the null hypothesis (H_0)**.

Why?

A p-value of 0.03 means that the observed difference would be relatively unlikely if the null hypothesis were true.

Since 0.03 is smaller than the 5% significance level, there is sufficient statistical evidence to reject H_0 .

Final Conclusion

Reject the null hypothesis at the 5% significance level.

This means the A/B test result shows a **statistically significant difference** between the two checkout page versions at the 5% level.

Important: Rejecting H_0 does not automatically mean that the new page is practically better. We also need to look at the direction and size of the improvement and the business metric being tested.

Decision Summary

Item	Value
p-value	0.03
Significance level (α)	0.05
Comparison	$0.03 < 0.05$

Decision	Reject H_0
Result	Statistically significant

Question 3

A/B Test Plan for Spotify

Business Question

Spotify wants to determine whether a **new playlist recommendation algorithm** increases the average daily listening time per user.

1. Hypotheses

Let:

- μ_1 = average daily listening time per user using the new algorithm
- μ_0 = average daily listening time per user using the old algorithm

Null Hypothesis (H_0):

$$H_0: \mu_1 \leq \mu_0$$

The new recommendation algorithm does not increase average daily listening time.

Alternative Hypothesis (H_1):

$$H_1: \mu_1 > \mu_0$$

The new recommendation algorithm increases average daily listening time.

2. Create Two Groups

Spotify can randomly divide eligible users into two groups:

Group	Algorithm	Purpose
Group A (Control)	Old recommendation algorithm	Baseline for comparison
Group B (Treatment)	New recommendation algorithm	Test the new algorithm

Random assignment helps make the two groups comparable and reduces selection bias.

3. Data to Collect

Spotify should collect data such as:

1. User ID or anonymous user identifier
2. Group assigned — A or B
3. Daily listening time in minutes
4. Number of songs played
5. Number of playlist sessions
6. Number of recommendation clicks
7. Date of activity

The **main metric** should be:

Average daily listening time per user (minutes/day).

For example:

Group	Number of Users	Average Daily Listening Time
A – Old Algorithm	1,000	80 minutes
B – New Algorithm	1,000	88 minutes

In this example, the new algorithm produces a higher average listening time.

4. Decision Rule

Choose a significance level such as:

$\alpha = 0.05$

After conducting the A/B test:

- If **p-value ≤ 0.05** , reject H_0 .
- If **p-value > 0.05** , do not reject H_0 .

5. How to Decide Whether the New Algorithm Is Better

The new algorithm can be considered statistically better if:

1. The new algorithm has a **higher average daily listening time**, and
2. The statistical test gives **p-value ≤ 0.05** .

Example

Suppose:

- Old algorithm average = **80 minutes/day**
- New algorithm average = **88 minutes/day**
- p-value = **0.02**

Since:

$$0.02 < 0.05$$

we reject H_0 .

Conclusion

There is statistically significant evidence that the new playlist recommendation algorithm increases average daily listening time per user. Therefore, Spotify could consider rolling out the new algorithm, while also checking other important metrics such as user satisfaction, skips, retention, and engagement.

A/B Testing Flow

Users → Random Assignment → Group A / Group B → Collect Data → Compare Average Listening Time → Calculate p-value → Make Decision

Question 4

A/B Test Example: Instagram

Business Question

Instagram wants to determine whether a **new button design for creating a post** encourages more users to create posts.

A/B Test Setup

Instagram can show:

- **Version A:** Existing post-creation button
- **Version B:** New post-creation button

Users are randomly divided into the two groups.

Null Hypothesis (H_0)

H_0 : The new button design does not increase the post-creation rate.

In statistical form:

$$H_0: p_{\square} \leq p_o$$

where:

- $p_{\square}$ = post-creation rate with the new button
- p_o = post-creation rate with the old button

Alternative Hypothesis (H_i)

H_i : The new button design increases the post-creation rate.

In statistical form:

$$H_i: p_{\square} > p_o$$

Metric to Measure

The main metric would be:

Post-Creation Rate

$$\frac{\text{Number of users who create a post}}{\text{Total number of users exposed to the design}} \times 100$$

Example

Suppose Instagram tests both versions with 1,000 users.

Version	Users	Users Who Created a Post	Post-Creation Rate
A – Old Button	1,000	100	10%
B – New Button	1,000	130	13%

The new design has a higher post-creation rate.

Instagram would then perform a statistical significance test.

If the resulting **p-value ≤ 0.05** , Instagram would reject H_0 and conclude that the new design has statistically significant evidence of increasing post creation.